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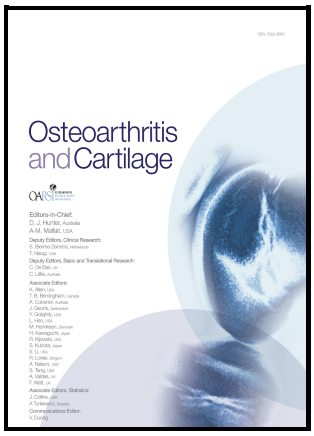

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ABSTRACT

Objective: To explore associations between hip muscle strength and cartilage defects (presence and severity) on magnetic resonance imaging (MRI) in young adults with hip/groin pain participating in sub-elite football.

Design: Sub-elite football players with hip/groin pain (> 6 months) completed assessments of isometric hip strength and functional task performance. Hip cartilage defects were assessed using the Scoring Hip Osteoarthritis with MRI (SHOMRI) tool. This exploratory, cross-sectional study used logistic and negative binomial models to assess the relationships between hip muscle strength or functional task performance and hip cartilage defects, controlling for body mass index, age, testing site and cam morphology, incorporating sex-specific interaction terms.

Results: One hundred and eighty-two (37 women) sub-elite (soccer or Australian football) players with hip/groin pain (age 26±7 years) were included. Greater hip extension strength was associated with higher cartilage total score (adjusted incidence rate ratio [aIRR] 1.01, 95%CI: 1.0 to 1.02, p=0.013) and superolateral hip cartilage defects (adjusted odds ratio (aOR) 1.03, 95% confidence interval (CI): 1.01 to 1.06, p<0.01). In female sub-elite football players, greater hip external rotation strength was associated with lateral cartilage defects (aOR 1.61, 95%CI: 1.05 to 2.48, p=0.03) and higher cartilage total score (aIRR 1.25, 95%CI: 1.01 to 1.66, p=0.042). A one-repetition increase in one-leg rise performance was related to lower odds of superomedial cartilage defects (aOR 0.96, 95%CI: 0.94 to 0.99, p<0.01).

Conclusions: Overall, there were few associations between peak isometric hip muscle strength and overall hip cartilage defects. It is possible that other factors may have relevance in sub-elite football players. Additional studies are needed to support or refute our findings that higher one leg rise performance was associated with reduced superomedial cartilage defect severity and greater hip extension strength was related to higher cartilage defect severity scores.

Keywords:

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Hip joint; Cartilage; muscle strength; function; Magnetic Resonance Imaging

INTRODUCTION

Participation in elite football codes is associated with hip osteoarthritis (OA) development in men but associations in women are not well understood[1, 2]. Cartilage damage is an important feature of early OA, particularly because once present, is mostly irreversible[3]. Fifty percent of football players already have hip cartilage defects by the third decade of life[4], which may explain the link between high-impact sport and hip OA. Identifying factors associated with cartilage defects will aid our understanding of OA disease in high-impact athletes.

Muscle strengthening is used in the management of hip OA. The link between muscle weakness and knee structural disease development[5] and progression[6] is established, but its role in hip OA is unclear. Hip muscle strength may play a role in hip cartilage loading as hip muscle forces can contribute up 95% of hip joint load during gait[7]. While there is limited research examining the relationship between peak strength and hip joint loads, reduced hip contact forces have been identified in people with symptomatic hip osteoarthritis, and were attributed to reduced lower limb muscle force production[8].

Functional performance measures are related to hip strength[9] but not synonymous, as they can reflect whole body movements elemental to sport and daily living activities, as well as confidence in the symptomatic hip [10, 11]. Better functional task performance appears to be important for halting knee OA progression[12, 13], but its role in hip OA has yet to be clarified.

Differences in hip muscle strength, functional task performance and cartilage health have been identified between sexes. Men with long-standing hip pain have greater hip muscle strength, functional task performance and cartilage defect severity than women[4, 14, 15]. To develop interventions to slow OA disease we must first understand if the relationship between hip muscle strength, functional task performance and cartilage defects/damage differs between sexes.

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Exploring the associations of potentially modifiable physical factors (such as strength/function) with hip joint cartilage defects (including sex differences) in the large Femoracetabular impingement and hip OsteoaRthritis Cohort (FORCe) will generate hypotheses for future studies. Our exploratory study of football players with hip/groin pain had three aims: i) examine if hip strength and/or functional task performance were associated with hip cartilage defects (presence or severity); ii) examine if such associations varied according to anatomical subregions of the hip; and iii) investigate if the association between strength/functional task performance and cartilage defects was stronger in men.

METHODS

Study design and participants

This cross-sectional study included all available baseline data from the FORCe study, an established cohort of 184 sub-elite soccer and Australian football (AF) players with hip/groin pain (including 38 women) recruited between August 2015 and October 2018 in Melbourne and Brisbane, Australia[16].

Eligibility criteria are described in Data supplement 1. Briefly, eligible participants had football-specific movement pain of gradual onset, with average pain between 4-7 on an 11-point numeric rating scale (NRS). In symptomatic participants, each hip was classified as either: i) symptomatic (positive flexion-adduction-internal-rotation (FADIR) test and hip/groin pain) or ii) other (no hip /groin or hip/groin pain and negative FADIR). For the present study we included symptomatic hips (n=288), with some participants reporting bilateral symptoms (n=107) and some unilateral (n=74) pain.

The parent FORCe study and the current data analyses were approved by both the La Trobe University (HEC 15-019) and University of Queensland (2015000916) Human Ethics Committees. All participants provided written informed consent. The current manuscript was written according to Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines[17].

Imaging

All participants underwent an unenhanced 3.0T MRI (Phillips Ingenia, The Netherlands). A 32-channel torso coil was placed over the hips and pelvis, with right and left hips imaged independently. Participants were positioned in supine using positioning aids to maintain hip internal rotation and neutral abduction/adduction. MRI scans included the following sequences: coronal proton density (PD) spectral attenuated inversion recovery (SPAIR), sagittal PD SPAIR and oblique axial PD SPAIR (Data supplement 2) which were evaluated by two musculoskeletal radiologists. The Scoring Hip Osteoarthritis with MRI (SHOMRI) scoring tool was completed independently by two musculoskeletal radiologists (FG, JL) who were blinded to participant demographics and clinical findings, with the final cartilage defect score determined by consensus. If disagreement occurred, a consensus read was completed in the presence of a third radiologist (TML).

SHOMRI cartilage scoring

Hip cartilage defects were assessed with the SHOMRI scoring system — an instrument with established reliability and validity for assessing presence and severity of cartilage defects[18]. A three-point scale is used to score cartilage defects: 0=no defects, 1= partial-thickness defects and 2=full-thickness defects in six femoral and four acetabular articular subregions[19].

To achieve our first aim, cartilage defects were scored as present if a partial or full thickness defect was identified in one or more femoral (n=6) or acetabular subregion (n=4) (Figure 1). For cartilage defect severity, each of the 10 subregions (0-2) were combined to determine an overall cartilage score (0-20). For our second aim, we combined adjacent femoral and acetabular cartilage subregions (e.g., superolateral femoral and superolateral acetabular were combined into superolateral cartilage) to create four combined femoroacetabular (superolateral, superomedial, anterior, posterior) subregions. In total, this provided us with 4 combined and two femoral (inferior and lateral) cartilage subregions. For each subregion, the presence (grade 1 or 2 cartilage defect in femoral or acetabular subregion) and severity (0-4 for combined femoroacetabular and 0-2 for

femoral only subregions) of cartilage defects was then determined. Intra-rater reproducibility was determined by each reader independently assessing 20 randomly selected images on two occasions, with readings completed two weeks apart. Inter-rater reliability was determined by each reader independently completing 50 images[20].

Insert Figure 1 here. Hip magnetic resonance images with subregion illustration.

Hip strength assessment

Bilateral hip flexion, extension, abduction, adduction, external and internal rotation were assessed with 'make' tests in standardised positions using a hand-held dynamometer (JTECH medical Commander muscle tester™)[21] (Data supplement 3). Participants stabilised themselves by holding onto the examination table and performed one sub-maximal familiarisation test for each muscle group, followed by three maximum voluntary contractions. Each isometric muscle contraction was performed with verbal encouragement, for a 5-second period, with 5 seconds break provided between each of the three trials. Thirty seconds break was provided prior to testing the next muscle group. Testers instructed participants when to commence muscle contraction using previously advocated standardised verbal encouragement "go ahead and push-push-push-push-relax"[21]. The highest of the three maximal voluntary contractions was used for analysis. Intra-rater reliability for this hip muscle strength testing protocol has been established with ICCs ranging from 0.76 to 0.98[21, 22]. Participants rated their pain during testing for each movement on an 11-point numeric rating scale.

Functional tasks

Functional task performance tasks were standardised and performed bilaterally with a 5 minute rest between tests[16]. Each test has been described in detail previously[16] and considered to be reliable

and clinically relevant for this population[23]. Briefly, the three functional tasks (described in Data supplement 3) included:

One-leg rise: participants began seated with the test limb hip and knee flexed to 90° and contralateral limb in 45° knee flexion not touching the ground and were instructed to stand upright from sitting (and return to sitting) as many times as possible, with arms crossed over the chest, in time with a metronome at a speed of 45 repetitions per minute.

Side-bridge endurance: participants commenced in side-lying, propped on the elbow with the test limb down, hips and knees extended, and feet stacked. Participants were asked to raise their pelvis and hold a straight trunk and lower limb position for as long as possible.

Hop for distance: participants placed their hands behind their backs and the toe of the test limb at the start of a measuring tape fixed to the floor. They were asked to hop as far as possible and maintain their balance for 3 seconds. Three successful trials were recorded after a practice trial, with the maximum trial used for analysis.

Subtle differences in the performance of flexion strength and hop for distance measurement existed between sites. During the flexion strength test, Melbourne participants held on to the plinth, but Brisbane participants had arms folded across their chest. Melbourne participants had hands behind backs during hop for distance task trials, while Brisbane participants did not. Additionally, assessors were unique at each testing venue, which may have implications for strength testing, so test site was included as a covariate within all multivariable analyses.

Demographic and patient-reported outcome measures

Demographic information (age, sex, height, mass, football code participation, training/competition frequency) was recorded and the International Hip Outcome Tool-33 (iHOT-33)[24] and the Copenhagen Hip and Groin Outcome Score (HAGOS)[25] completed by each participant. The iHOT-33 and the HAGOS quantify hip/groin pain burden across 4 and 6 subscales, respectively [24, 25]. A higher score (0-100) represents better QOL. PROMs were used to describe the hip-related burden of the

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cohort and are validated and recommended for young adults with hip/groin pain[26, 27]. Cam morphology was assessed by one observer (JJH) blinded to hip classification, from standardised radiographs using quantitative (alpha angle) methods[28].

Statistical analysis

Data analyses were performed using StataCorp. 2023. *Stata Statistical Software: Release 18*. College Station, TX: StataCorp LLC. Patient and hip demographic characteristics were stratified by sex and reported as frequencies and median and interquartile range (IQR) or mean ± standard deviation (SD). Intra- and inter-rater reliability was determined with weighted kappa. For the first and second study aim, binomial logistic regression models using generalised estimating equations (GEE) to control for within-person correlation between right and left hip outcomes were used to determine the association between hip muscle strength or functional task performance, and cartilage defects (presence), with findings reported as odds ratios (OR) with associated 95% confidence intervals and P-values. Negative binomial regression models with GEE were used to estimate the association between hip strength or functional task performance and the severity of cartilage defects, with findings reported as incidence rate ratios (IRR) with associated 95% confidence intervals and P-values. For aim three, an interaction term (hip strength/function *sex) was included in each regression model to test if the association between strength/functional task performance and cartilage defects was stronger in men. If the interaction analysis indicated the association was modified by sex, data from men and women were stratified and analyses adjusted for test site (Melbourne, Brisbane), age, BMI and cam morphology (alpha angle >60°). If not modified by sex, data from females and males were pooled and all analyses adjusted for sex, test site, age, BMI and cam morphology. For interpretability, strength OR and IRR analyses for 1 Newton (N) force increases were converted to 10 N to approximate 1kg increase in force.

RESULTS

The 182 symptomatic football players included 38 women (59 hips) with median symptom duration of 24 months, and 146 men (229 symptomatic hips) with median symptom duration of 30 months (Table 1). Most participants exercised intensely (breathless, sweating) for more than 60 minutes, 2-3 times per week. The median iHOT-33 score was 61 and 63 in women and men, respectively. The median pain score reported during muscle testing was zero for all muscle groups (Data supplement 4). Three of the most prevalent sites for cartilage defects in all (n=288) symptomatic hips, were the superolateral (41%), lateral femoral (29%) and superomedial (22%) subregions. Of 167 (58%) hips with a cartilage defect, 101 (35%) hips had a defect in two or more subregions. (Figure 1). Alpha angle data were missing for 14 participants, so data from 274 participants were used for GEE regression analyses.

Insert Table 1. Demographic characteristics of high impact athletes with hip/groin pain.

Reliability

Intra-rater (FG, 0.86; 95%CI: 0.73 to 1.00/JL, 0.94; 95%CI: 0.80 to 1.00) and inter-rater reliability (0.96; 95%CI 0.88 to 1.00) was almost perfect for cartilage defect scoring[29].

Aim 1. Association between hip strength and functional task performance and hip cartilage defects (presence or severity).

There was one association between (10 N) increased hip extension strength and increased hip cartilage defect severity (IRR=1.01, 95%CI:1.0 to 1.02, p=0.013) and no associations between functional task performance measures and the presence of any hip cartilage defect, or overall defect severity (Table 2). Unadjusted analyses are presented in Data supplement 5.

Insert Table 2. Associations of hip strength or functional task performance with hip cartilage defect (presence and severity) in sub-elite football players (n=274 hips).

Aim 2. Association of hip strength and functional task performance with the presence or severity of cartilage defects in specific anatomical subregions.

There were no associations between hip strength measures and the presence of a cartilage defect in the superolateral or superomedial regions (Table 3) (Data supplement 6). Greater extension strength (10 N increase) was related to the severity of a superolateral cartilage defect (IRR=1.03, 95%CI:1.00 to 1.06, p<0.01). Increased one-leg rise repetitions were associated with a lower odds of a superomedial cartilage defect (OR=0.96, 95%CI:0.94 to 0.99, p<0.01). Unadjusted analyses are presented in Data supplement 6.

Insert Table 3. Associations of hip strength and functional task performance with hip cartilage subregion defect (presence and severity) in sub-elite football players (n=274 hips).

Aim 3. Interaction between strength and/or functional task performance and sex.

Only two sex-dependent associations were found (Tables 2 and 3). Increased external rotation strength (of 10 N) was positively associated with overall cartilage defect severity (IRR=1.25, 95%CI:1.01 to 1.56, p=0.042) and the presence of a lateral femoral cartilage defect (OR= 1.61, 95%CI:1.05 to 2.48, p=0.03) in women (p=0.043 and p=0.01, respectively) but not men.

DISCUSSION

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This study explored the association between physical measures (hip muscle strength and functional task performance) and cartilage defects in sub-elite football players with hip/groin pain. Overall, few associations were found, suggesting that the physical measures examined may not be as important as other potential factors (e.g., male sex, cam morphology, hip dysplasia, age[30, 31]) associated with hip cartilage defects in football players. Higher hip extension strength was associated with greater odds of overall and superolateral cartilage defect severity. When exploring subregional defects, we found a modest association between higher one-leg rise repetitions and decreased superolateral cartilage defect severity. Two sex-specific associations were found. Greater external rotation strength was associated with increased cartilage defect severity (any location) and the presence of lateral femoral cartilage defects in women only.

Despite the generally weak associations between strength/functional task performance and cartilage defects, our findings suggest that poor one-leg rise performance may be a useful clinical tool to indicate possible region specific (superomedial) cartilage defects in high-impact athletes, however more studies are needed to support or refute our findings. Consistent with our findings, poorer one-leg rise performance has also been reported in individuals with chondropathy at the time of hip arthroscopy[15]. Fewer one-leg rise repetitions suggests weaker lower-limb (i.e., including hip) muscle strength[32], however only weak to moderate associations have been shown between functional task performance and hip strength[33, 34]. Extrapolating from studies of knee OA, peak quadriceps strength measures have a moderate correlation with one-leg rise performance, but the association may diminish in higher-functioning individuals[32]. As a composite test, and one that is more challenging than isolated strength testing, one-leg rise may be more sensitive for detecting early hip OA imaging signs in young adult athletes than unidimensional tests (such as isometric strength). Unlike the knee[12, 13], associations between impaired hip functional task performance and progression of hip cartilage defects has not been evaluated. Future studies could explore the association of one-leg rise with longitudinal hip cartilage change data.

We found that a 10 N increase in hip extensor strength was related to small increases (1% and 3%) in total joint and superolateral cartilage defect severity, respectively, however the most notable finding of our study was the lack of association between most clinical peak isometric strength measures and hip cartilage defects. This may be explained by the high-functioning cohort we examined (i.e., playing sport for more than an hour at least twice per week) and the absence of radiographic OA. The clinical relevance of this modest relationship is difficult to determine, as it remains unclear how greater extensor strength is related to worse cartilage health and may be due to random factors. Existing investigations of those seeking surgery for hip/groin pain also found no association of cartilage defect severity with isokinetic abductor or adductor muscle strength[35]. Yet studies using advanced MRI measures of cartilage composition have reported that increased T2 or T1rho relaxation times (indicating poorer cartilage quality) was related to smaller hip abductor and flexor muscle forces during walking [36]. Further, in older men, hip cartilage defects identified using MRI were associated with combined isometric hip and knee extension weakness[37]. It is possible that those with stronger hip extension perform more high-load activities, thus increasing the likelihood of a cartilage defect, although this cannot be confirmed in our cross-sectional study. The relationship between hip muscle strength and cartilage health remains unclear and requires confirmation in high-quality longitudinal studies.

In women, we found that a 10 N increase in external rotation strength was associated with a 39% increased risk of a lateral femoral cartilage defect and a 75% increased risk of more severe cartilage defects overall. It is important to note that we previously reported decreased hip adduction, external and internal rotation strength in symptomatic football athletes compared to the control group[38]. Therefore, the clinical implication of these findings are difficult to interpret as women football athletes with hip/groin pain are likely to have weaker external rotation than their pain free peers. Hip external rotator and extensor muscles are important for sprinting and cutting (i.e., change of direction while running)[39]. Extensor muscles also produce the greatest torque across the hip joint[39] and hip muscle forces are thought to contribute more to joint contact forces with

increased running speed than ground reaction forces[40]. Athletic activities, such as sprinting and cutting, combined with impingement may contribute to hip/groin pain development and hip OA incidence[41-43]. However, it is unknown whether football players with stronger extensor muscles run faster or perform more cutting manoeuvres leading to a higher likelihood of cartilage defects. In contrast, it is also possible that those with stronger muscles have higher joint compression, which could mean higher joint stability that might be beneficial for the joint and its structures. Additionally, women perform cutting manoeuvres with more hip internal and external rotation range of motion and less hip (and knee) flexion, particularly when fatigued, compared to men[44] which may explain our finding of a relationship between external rotation strength and cartilage defects in women, but not men. It is possible that sex may alter the association of hip external rotation strength and cartilage defects reflecting the importance of examining differences in men and women athletes[45].

The clinical implications of our findings indicate that overall isometric hip muscle strength does not appear to be associated with hip cartilage health in high-impact athletes. Until this relationship is clarified in longitudinal studies (that can indicate casuation), hip muscle strengthening should remain a key component of rehabilitation programs for athletes with hip/groin pain[46]. Multi-dimensional hip functional task performance (such as one-leg rise) may also be important treatment targets for hip/groin pain conditions to improve symptoms and quality of life[47]. The promotion of general physical activity for overall joint health should also be prioritised in rehabilitation programs. Longitudinal studies are needed to understand if changes to hip muscle strength and other physical impairments can modify hip cartilage structural degradation.

Strengths and limitations

The strengths of our study include the large sample (n=182/288 hips) of football players with long-standing hip/groin pain[16]. By creating combined cartilage regions from adjacent femoral and acetabular SHOMRI subregions we overcame reported variability associated with separate analyses of already thin hip cartilage[18, 48]. The dichotomous SHOMRI outcomes used in this study may have

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left us underpowered for some analyses. Similarly, the sensitivity of the SHOMRI cartilage score as a simplified rating to facilitate time efficient completion of the overall assessment tool for all joint features, may not be sensitive to cartilage defect severity[19]. The proportion of women was also lower than men and so this study may have been underpowered to detect sex interaction effects and then perform stratified analyses. We conducted a number of regression analyses that may have increased the risk of a Type I error, however adjustment for such analyses is not without its own limitations, including increased probability of Type II errors[49]. Although participants had minimal pain during strength tests, which aligns with pain outcomes in other studies of soccer players with long-standing hip pain[50] and individuals awaiting arthroscopy for FAI syndrome[51], it is possible that pain severity during testing may have influenced muscle strength and functional performance outcomes. We did not include acetabular morphology radiographic measures as a covariate due to the low prevalence of variations in acetabular morphology in the cohort (7% for pincer and 4% for hip dysplasia)[28]. It is possible however, that we underestimated acetabular morphology prevalence as we did not include a second radiographic criteria (e.g., focal acetabular retroversion determined by the crossover sign) due to poor reproducibility of radiographic assessment[52]. We were interested in the relationship between muscle strength/function as the exposure variable, but due to cross-sectional study design, we acknowledge that our results may be interpreted the same way with cartilage as the exposure variable. It is also unclear if findings can be generalised to athletes other than football players or to less active individuals with long-standing hip/groin pain.

CONCLUSIONS

In young adult football players with hip/groin pain, it appears that isometric hip muscle strength or functional task performace may not be as important as other factors associated with the presence or severity of hip cartilage defects cartilage defects in sub-elite football players. The one-leg rise task may provide a clinical indicator of superomedial cartilage defect (presence and severity) and sex

could alter the association of hip external rotation strength and cartilage defects reflecting the importance of examining differences in men and women athletes. Due to the cross-sectional study design, we cannot establish causation but these findings may be useful for hypothesis generation in longitudinal studies that explore the relationship of physical features (such as muscle strength and functional task performance) with longitudinal hip cartilage changes or the effects of rehabilitation interventions on the development or progression of hip cartilage defects.

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Author contributions:

SC, JH, JK, KC, AC, SW contributed to conception and design of the study, analysis and interpretation of data, drafting and revising of the manuscript. JH, MK, MS, PL, JL, FG and TL contributed to collection and assembly of data and critical revision of the final manuscript and JH, JK, KC, AC, SW, and MK provided statistical expertise and MK assisted with analysis and interpretation of data. KC also obtained funding for the FORCe study and provided final approval of the article.

Competing interests.

The authors declare they have no competing interests.

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Figure legends

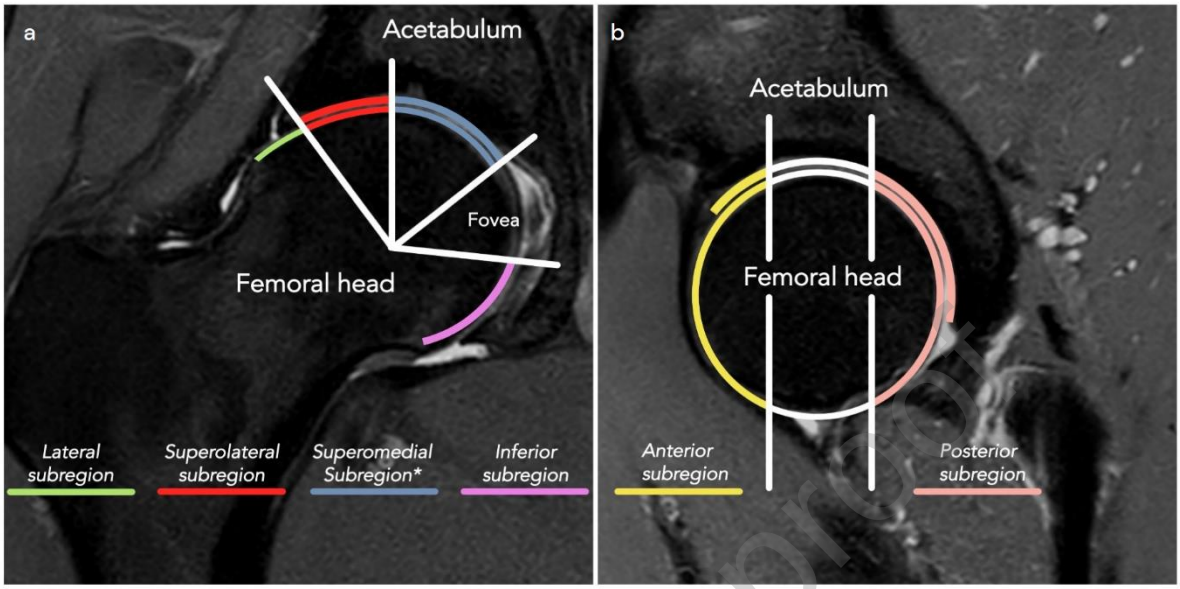

Figure 1. Hip magnetic resonance images with subregion illustration; a-coronal MRI evaluation of femoral and acetabular cartilage in the superolateral and superomedial subregions and femoral cartilage in the lateral and inferior subregions, b-sagittal MRI evaluation of femoral and acetabular cartilage in the anterior and posterior subregions.

REFERENCES

1. Vigdorchik JM, Nepple JJ, Eftekhary N, Leunig M, Clohisy JC. What is the association of elite sporting activities with the development of hip osteoarthritis? The American journal of sports medicine. 2017;45(4):961-4.

2. Lohkamp M, Kromer TO, Schmitt H. Osteoarthritis and joint replacements of the lower limb and spine in ex-professional soccer players: A systematic review. Scand J Med Sci Sports. 2017;27(10):1038-49. doi: 10.1111/sms.12846.

3. Ding C, Garnero P, Cicuttini F, Scott F, Cooley H, Jones G. Knee cartilage defects: association with early radiographic osteoarthritis, decreased cartilage volume, increased joint surface area and

Journal Pre-proof

type II collagen breakdown. Osteoarthritis Cartilage. 2005;13(3):198-205. doi: 10.1016/j.joca.2004.11.007.

4. Heerey JJ, Srinivasan R, Agricola R, Smith A, Kemp JL, Pizzari T, et al. Prevalence of early hip OA features on MRI in high-impact athletes. The femoroacetabular impingement and hip osteoarthritis cohort (FORCe) study. Osteoarthritis Cartilage. 2021;29(3):323-34. doi: 10.1016/j.joca.2020.12.013.

5. Øiestad BE, Juhl CB, Culvenor AG, Berg B, Thorlund JB. Knee extensor muscle weakness is a risk factor for the development of knee osteoarthritis: an updated systematic review and meta-analysis including 46 819 men and women. Br. J. Sports Med. 2022;56(6):349-55.

6. Patterson BE, Girdwood MA, West TJ, Bruder AM, Oiestad BE, Juhl C, et al. Muscle strength and osteoarthritis of the knee: a systematic review and meta-analysis of longitudinal studies. Skeletal Radiol. 2023;52(11):2085-97. doi: 10.1007/s00256-022-04266-4.

7. Correa TA, Crossley KM, Kim HJ, Pandy MG. Contributions of individual muscles to hip joint contact force in normal walking. J. Biomech. 2010;43(8):1618-22. doi: 10.1016/j.jbiomech.2010.02.008.

8. Diamond L, Hoang H, Barrett R, Loureiro A, Constantinou M, Lloyd D, et al. Individuals with mild-to-moderate hip osteoarthritis walk with lower hip joint contact forces despite higher levels of muscle co-contraction compared to healthy individuals. Osteoarthritis Cartilage. 2020;28(7):924-31.

9. Kemp JL, Schache AG, Makdissi M, Sims KJ, Crossley KM. Greater understanding of normal hip physical function may guide clinicians in providing targeted rehabilitation programmes. Journal of Science and Medicine in Sport. 2013;16(4):292-6. doi: https://doi.org/10.1016/j.jsams.2012.11.887.

10. Kivlan BR, Martin RL. Functional performance testing of the hip in athletes: a systematic review for reliability and validity. Int J Sports Phys Ther. 2012;7(4):402-12.

11. Reiman MP, Manske RC. The assessment of function: How is it measured? A clinical perspective. Journal of Manual & Manipulative Therapy. 2011;19(2):91-9.

Journal Pre-proof

12. Patterson B, Culvenor AG, Barton CJ, Guermazi A, Stefanik J, Morris HG, et al. Poor functional performance 1 year after ACL reconstruction increases the risk of early osteoarthritis progression. Br J Sports Med. 2020;54(9):546-53. doi: 10.1136/bjsports-2019-101503.

13. Thorstensson CA, Petersson IF, Jacobsson LT, Boegard TL, Roos EM. Reduced functional performance in the lower extremity predicted radiographic knee osteoarthritis five years later. Ann Rheum Dis. 2004;63(4):402-7. doi: 10.1136/ard.2003.007583.

14. Bizzini M, Schaub G, Ferrari E, Monn S, Leunig M, Casartelli NC, et al. Hip muscle strength in male and female patients with femoroacetabular impingement syndrome: Comparison to healthy controls and athletes. Phys Ther Sport. 2023;61:142-8. doi: 10.1016/j.ptsp.2023.03.010.

15. Kemp JL, Risberg MA, Schache AG, Makdissi M, Pritchard MG, Crossley KM. Patients With Chondrolabral Pathology Have Bilateral Functional Impairments 12 to 24 Months After Unilateral Hip Arthroscopy: A Cross-sectional Study. Journal of Orthopaedic & Sports Physical Therapy. 2016;46(11):947-56. doi: 10.2519/jospt.2016.6577.

16. Crossley KM, Pandy MG, Majumdar S, Smith AJ, Agricola R, Semciw AI, et al. Femoroacetabular impingement and hip OsteoaRthritis Cohort (FORCe): protocol for a prospective study. J Physiother. 2018;64(1):55. doi: 10.1016/j.jphys.2017.10.004.

17. Vandenbroucke JP, Von Elm E, Altman DG, Gøtzsche PC, Mulrow CD, Pocock SJ, et al. Strengthening the Reporting of Observational Studies in Epidemiology (STROBE): explanation and elaboration. PLoS medicine. 2007;4(10):e297.

18. Neumann J, Zhang AL, Schwaiger BJ, Samaan MA, Souza R, Foreman SC, et al. Validation of scoring hip osteoarthritis with MRI (SHOMRI) scores using hip arthroscopy as a standard of reference. European Radiology. 2019;29(2):578-87. doi: 10.1007/s00330-018-5623-8.

19. Lee S, Nardo L, Kumar D, Wyatt CR, Souza RB, Lynch J, et al. Scoring hip osteoarthritis with MRI (SHOMRI): A whole joint osteoarthritis evaluation system. J Magn Reson Imaging. 2015;41(6):1549-57. doi: 10.1002/jmri.24722.

20. Heerey J, Souza, RB, Gassert, F, Luitjens J, Kemp, JL, Scholes, MS, Crossley, KM, Link, T. The Scoring of Hip Osteoarthritis with MRI: definitions for reporting prevalence, severity and progression. Skeletal Radiology 2023.

Journal Pre-proof

21. Thorborg K, Petersen J, Magnusson SP, Hölmich P. Clinical assessment of hip strength using a hand-held dynamometer is reliable. Scandinavian Journal of Medicine & Science in Sports. 2010;20(3):493-501. doi: 10.1111/j.1600-0838.2009.00958.x.

22. Thorborg K, Bandholm T, Holmich P. Hip- and knee-strength assessments using a hand-held dynamometer with external belt-fixation are inter-tester reliable. Knee Surg Sports Traumatol Arthrosc. 2013;21(3):550-5. doi: 10.1007/s00167-012-2115-2.

23. Mosler AB, Kemp J, King M, Lawrenson PR, Semciw A, Freke M, et al. Standardised measurement of physical capacity in young and middle-aged active adults with hip-related pain: recommendations from the first International Hip-related Pain Research Network (IHiPRN) meeting, Zurich, 2018. Br J Sports Med. 2020;54(12):702-10. doi: 10.1136/bjsports-2019-101457.

24. Mohtadi NG, Griffin DR, Pedersen ME, Chan D, Safran MR, Parsons N, et al. The Development and validation of a self-administered quality-of-life outcome measure for young, active patients with symptomatic hip disease: the International Hip Outcome Tool (iHOT-33). Arthroscopy. 2012;28(5):595-605; quiz 6-10 e1. doi: 10.1016/j.arthro.2012.03.013.

25. Thorborg K, Holmich P, Christensen R, Petersen J, Roos EM. The Copenhagen Hip and Groin Outcome Score (HAGOS): development and validation according to the COSMIN checklist. Br J Sports Med. 2011;45(6):478-91. doi: 10.1136/bjsm.2010.080937.

26. Thorborg K, Tijssen M, Habets B, Bartels EM, Roos EM, Kemp J, et al. Patient-Reported Outcome (PRO) questionnaires for young to middle-aged adults with hip and groin disability: a systematic review of the clinimetric evidence. Br J Sports Med. 2015;49(12):812. doi: 10.1136/bjsports-2014-094224.

27. Scholes MJ, King MG, Crossley KM, Jones DM, Semciw AI, Mentiplay BF, et al. The Validity, Reliability, and Responsiveness of the International Hip Outcome Tool-33 (iHOT-33) in Patients With Hip and Groin Pain Treated Without Surgery. Am J Sport Med. 2021;49(10):2677-88. doi: 10.1177/03635465211027180.

28. Heerey J, Agricola R, Smith A, Kemp J, Pizzari T, King M, et al. The Size and Prevalence of Bony Hip Morphology Do Not Differ Between Football Players With and Without Hip and/or

Journal Pre-proof

Groin Pain: Findings From the FORCe Cohort. J Orthop Sports Phys Ther. 2021;51(3):115-25. doi: 10.2519/jospt.2021.9622.

29. Landis JR, Koch GG. The measurement of observer agreement for categorical data. Biometrics. 1977;33(1):159-74.
30. Heerey J, Kemp J, Agricola R, Srinivasan R, Smith A, Pizzari T, et al. Cam morphology is associated with MRI-defined cartilage defects and labral tears: a case-control study of 237 young adult football players with and without hip and groin pain. BMJ Open Sport Exerc Med. 2021;7(4):e001199. doi: 10.1136/bmjsem-2021-001199.
31. Ishoi L, Thorborg K, Kraemer O, Lund B, Mygind-Klavsen B, Holmich P. Demographic and Radiographic Factors Associated With Intra-articular Hip Cartilage Injury: A Cross-sectional Study of 1511 Hip Arthroscopy Procedures. Am J Sports Med. 2019;47(11):2617-25. doi: 10.1177/0363546519861088.
32. West TJ, Bruder AM, Crossley KM, Girdwood MA, Scholes MJ, To LK, et al. Does the one-leg rise test reflect quadriceps strength in individuals following anterior cruciate ligament reconstruction? Phys Ther Sport. 2023;63:104-11. doi: 10.1016/j.ptsp.2023.07.008.
33. Thongchoomsin S, Bovonsunthonchai S, Joseph L, Chamnongkich S. Clinimetric properties of the one-leg sit-to-stand test in examining unilateral lower limb muscle strength among young adults. Int J Clin Pract. 2020;74(9):e13556. doi: ARTN e1355610.1111/ijcp.13556.
34. Crossley KM, Zhang WJ, Schache AG, Bryant A, Cowan SM. Performance on the single-leg squat task indicates hip abductor muscle function. Am J Sports Med. 2011;39(4):866-73. doi: 10.1177/0363546510395456.
35. Gomes D, Ribeiro DC, Canella RP, Ferreira T, da Costa GV, Okubo R, et al. Association between severity of hip chondrolabral injuries, dynamic hip muscle strength and quality of life: A cross-sectional study in patients with femoroacetabular impingement syndrome scheduled for hip arthroscopy. Clinical Biomechanics. 2021;84:105348. doi: ARTN 105348

10.1016/j.clinbiomech.2021.105348.

Journal Pre-proof

36. Samaan MA, Zhang AL, Popovic T, Pedoia V, Majumdar S, Souza RB. Hip joint muscle forces during gait in patients with femoroacetabular impingement syndrome are associated with patient reported outcomes and cartilage composition. J Biomech. 2019;84:138-46. doi: 10.1016/j.jbiomech.2018.12.026.

37. Ahedi HG, Aitken DA, Blizzard LC, Ding CH, Cicuttini FM, Jones G. Correlates of Hip Cartilage Defects: A Cross-sectional Study in Older Adults. J. Rheumatol. 2016;43(7):1406-12. doi: 10.3899/jrheum.151001.

38. Bonello C, King MG, Crossley KM, Heerey JJ, Scholes MJ, Lawrenson P, et al. The association between hip/groin pain and hip strength in football players: An exploratory analysis of the FORCe cohort. Journal of Science and Medicine in Sport. 2023;26(9):471-5. doi: 10.1016/j.jsams.2023.07.008.

39. Neumann DA. Kinesiology of the hip: a focus on muscular actions. J Orthop Sports Phys Ther. 2010;40(2):82-94. doi: 10.2519/jospt.2010.3025.

40. Giarmatzis G, Jonkers I, Wesseling M, Van Rossom S, Verschueren S. Loading of Hip Measured by Hip Contact Forces at Different Speeds of Walking and Running. Journal of Bone and Mineral Research. 2015;30(8):1431-40. doi: 10.1002/jbmr.2483.

41. Clohisy JC, Baca G, Beaule PE, Kim YJ, Larson CM, Millis MB, et al. Descriptive epidemiology of femoroacetabular impingement: a North American cohort of patients undergoing surgery. Am J Sports Med. 2013;41(6):1348-56. doi: 10.1177/0363546513488861.

42. Siebenrock KA, Ferner F, Noble PC, Santore RF, Werlen S, Mamisch TC. The cam-type deformity of the proximal femur arises in childhood in response to vigorous sporting activity. Clin Orthop Relat Res. 2011;469(11):3229-40. doi: 10.1007/s11999-011-1945-4.

43. van Klij P, Heerey J, Waarsing JH, Agricola R. The Prevalence of Cam and Pincer Morphology and Its Association With Development of Hip Osteoarthritis. J Orthop Sports Phys Ther. 2018;48(4):230-8. doi: 10.2519/jospt.2018.7816.

44. McGovern A, Dude C, Munkley D, Martin T, Wallace D, Feinn R, et al. Lower limb kinematics of male and female soccer players during a self-selected cutting maneuver: Effects of prolonged activity. Knee. 2015;22(6):510-6. doi: 10.1016/j.knee.2015.05.005.

Journal Pre-proof

45. Paul RW, Sonnier JH, Johnson EE, Hall AT, Osman A, Connors GM, et al. Inequalities in the Evaluation of Male Versus Female Athletes in Sports Medicine Research: A Systematic Review. Am J Sports Med. 2023;51(12):3335-42. doi: 10.1177/03635465221131281.

46. Kemp JL, Risberg MA, Mosler A, Harris-Hayes M, Serner A, Moksnes H, et al. Physiotherapist-led treatment for young to middle-aged active adults with hip-related pain: consensus recommendations from the International Hip-related Pain Research Network, Zurich 2018. Br J Sports Med. 2020;54(9):504-11. doi: 10.1136/bjsports-2019-101458.

47. Kemp JL, Mosler AB, Hart H, Bizzini M, Chang S, Scholes MJ, et al. Improving function in people with hip-related pain: a systematic review and meta-analysis of physiotherapist-led interventions for hip-related pain. Br J Sports Med. 2020;54(23):1382-94. doi: 10.1136/bjsports-2019-101690.

48. Roemer FW, Hunter DJ, Winterstein A, Li L, Kim YJ, Cibere J, et al. Hip Osteoarthritis MRI Scoring System (HOAMS): reliability and associations with radiographic and clinical findings. Osteoarthritis Cartilage. 2011;19(8):946-62. doi: 10.1016/j.joca.2011.04.003.

49. Perneger TV. What's wrong with Bonferroni adjustments. BMJ. 1998;316(7139):1236-8. doi: 10.1136/bmj.316.7139.1236.

50. Rafn BS, Tang L, Nielsen MP, Branci S, Hölmich P, Thorborg K. Hip strength testing of soccer players with long-standing hip and groin pain: what are the clinical implications of pain during testing? Clinical Journal of Sport Medicine. 2016;26(3):210-5.

51. Diamond LE, Wrigley TV, Hinman RS, Hodges PW, O'Donnell J, Takla A, et al. Isometric and isokinetic hip strength and agonist/antagonist ratios in symptomatic femoroacetabular impingement. J Sci Med Sport. 2016;19(9):696-701. doi: 10.1016/j.jsams.2015.10.002.

52. Clohisy JC, Carlisle JC, Trousdale R, Kim YJ, Beaule PE, Morgan P, et al. Radiographic Evaluation of the Hip has Limited Reliability. Clinical Orthopaedics and Related Research. 2009;467(3):666-75. doi: 10.1007/s11999-008-0626-4.

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Table 1. Demographic characteristics of high impact athletes with hip/groin pain.

Participant demographics (per person)	All	Women	Men
Number of participants (%)	182	37 (20%)	145 (80%)
Age (years)	26 [7]	25 [9]	27 [7]
Height (m)	1.8 [0.1]	1.7 [0.1]	1.8 [0.1]
Weight (kg)	78 [14.3]	63.1 [12.2]	79.8 [12.6]
Body mass index (kg/m^2)	24.2 [3.4]	22.7 [2.7]	24.5 [3.1]
Exercise frequency per week			
2-3 sessions	89%	95%	88%
≥4 sessions	11%	5%	12%
Symptom duration (months)	24 [31]	24 [25]	30 [42]
HAGOS scores			
HAGOS-Symptoms	61 [17]	57 [29]	61 [14]
HAGOS-Pain	75 [18]	72 [23]	75 [20]
iHOT33-Total	64 [24]	66 [24]	64 [24]
iHOT33-Q16 Pain	55 [32]	58 [32]	55 [31]
Participant demographics (per hip)	**All**	**Women**	**Men**
Number of symptomatic hips	288	59 (21%)	229 (79%)
Hip test site (%)	215 Mel (75%)	46 Mel (78%)	169 Mel (74%)
Alpha angle >60°	274 (67%)	23 (41% of women)	170 (78% of men)
Strength (N)/function			
Abduction	134 (35)	97 (24)	143 (31)
Adduction	128 (38)	93 (22)	138 (35)
Extension	272 (74)	222 (57)	285 (72)
External rotation	115 (29)	89 (18)	122 (27)
Internal rotation	104 (27)	84 (17)	109 (27)
Flexion	207 (69)	160 (39)	219 (69)
One-leg rise (repetitions)	19 (13)	20 (17)	19 (12)
Side-bridge (seconds)	81 (33)	68 (34)	84 (32)
Hop for distance (cm)	129 (29)	109 (17)	134 (30)

Per person data reported as count (%) and median [interquartile range].

Per hip demographic data reported as count (%) and strength/function data reported as mean (SD).

cm=Centimetres, HAGOS= Copenhagen Hip and Groin Outcome Score, iHOT33=International Hip Outcome Tool 33, kg=Kilograms, m=Metres, Mel=Melbourne test site; N=Newton.

Table 2. Association of hip strength or functional task performance with hip cartilage defect (presence and severity) in sub-elite football players (n=274 hips).

		Hips[a]	*sex[b]	Multivariable logistic regression[c]		
Variables	Unit	n	p	aOR	95%CI	p
Abduction	10 N	274	0.389	0.997	0.911 to 1.091	0.955
Adduction	10 N	274	0.723	1.024	0.944 to 1.112	0.564
Extension	10 N	274	0.412	1.009	0.988 to 1.031	0.409
External rotation	10 N	274	0.16	1.074	0.963 to 1.1974	0.202
Internal rotation	10 N	274	0.899	0.997	0.889 to 1.119	0.964
Flexion	10 N	274	0.776	1.025	0.975 to 1.078	0.336

One-leg rise	reps	274	0.595	0.987	0.964 to 1.01	0.26
Side-bridge	sec	274	0.559	0.997	0.987 to 1.006	0.468
Hop for distance	cm	274	0.326	1.007	0.995 to 1.018	0.261
		Hips[a]	*sex[b]	Multivariable negative binomial regression[c]		
Variables	Unit	n	p	aIRR	95%CI	p
Abduction	10 N	274	0.607	1.0	0.961 to 1.04	0.986
Adduction	10 N	274	0.712	1.029	0.988 to 1.071	0.164
Extension	10 N	274	0.482	**1.01**	**1.002 to 1.019**	**0.013**
External rotation		274	**0.043**			
Male	10 N	170		0.993	0.939 to 1.051	0.817
Female	10 N	23		**1.254**	**1.009 to 1.559**	**0.042**
Internal rotation	10 N	274	0.583	0.993	0.936 to 1.054	0.820
Flexion	10 N	274	0.415	1.013	0.99 to 1.036	0.27
One-leg rise	reps	274	0.462	0.993	0.981 to 1.005	0.222
Side-bridge	sec	274	0.488	0.999	0.994 to 1.004	0.704
Hop for distance	cm	274	0.567	1.003	0.997 to 1.009	0.303

Bold values indicate a statistically significant interaction or association (p<0.05)

For exposure variables (raw score in N, reps, sec, cm), an OR/IRR >1 represents increased odds of cartilage defect presence or severity, respectively. For example, a 10 N (approximately 1 kg) increase in adduction strength could be interpreted as having a 2% increased odds of a hip cartilage defect.

[a]Number of hips with radiological alpha angle data.

[b]Strength/function variable*sex interaction.

[c]Adjusted for sex, test site, age, body mass index, alpha angle >60°.

cm=Centimetres, CI=Confidence interval, aIRR=Adjusted incidence rate ration, N-Newton, aOR=Adjusted odds ratio, reps=Repetitions, sec=Second.

Table 3. Multivariable associations of hip strength or functional task performance with hip cartilage subregion defects (presence and severity) in sub-elite football players (n=274 hips).

			Multivariable binomial logistic regression[c]				Multivariable binomial logistic regression[c]				Multivariable binomial logistic regression[c]			
		Hips[a]	*sex[b]	aOR	95%CI	p	*sex[b]	aOR	95%CI	p	*sex[b]	aOR	95%CI	p
Variables	Unit	n	p	Any superolateral cartilage defect (n=118)			p	Any superomedial cartilage defect (n=64)			p	Any lateral femoral cartilage defect (n=83)		
Abduction	**10 N**	274	0.859	1.014	0.927 to 1.11	0.760	0.548	0.999	0.901 to 1.108	0.988	0.31	0.977	0.885 to 1.063	0.515
Adduction	**10 N**	274	0.926	1.041	0.958 to 1.13	0.345	0.648	1.045	0.949 to 1.152	0.370	0.444	0.982	0.903 to 1.067	0.667
Extension	**10 N**	274	0.177	1.039	0.998 to 1.083	0.062	0.638	1.012	0.978 to 1.047	0.486	0.998	0.989	0.968 to 1.01	0.309
External rotation	**10 N**	274	0.214	1.034	0.927 to 1.153	0.548	0.065	1.025	0.902 to 1.165	0.703	**0.01**			
Male	**10 N**	170										0.896	0.797 to 1.007	0.066
Female	**10 N**	23										**1.611**	**1.046 to 2.481**	**0.03**
Internal rotation	**10 N**	274	0.742	1.0	0.891 to 1.122	0.993	0.973	0.904	0.789 to 1.036	0.147	0.663	0.974	0.867 to 1.094	0.661
Flexion	**10 N**	274	0.953	1.032	0.981 to 1.085	0.222	0.196	1.002	0.943 to 1.063	0.960	0.497	1.01	0.96 to 1.062	0.699
One-leg rise	reps	274	0.659	1.001	0.977 to 1.026	0.92	0.452	**0.964**	**0.937 to 0.991**	**0.009**	0.942	0.991	0.968 to 1.016	0.496
Side-bridge	sec	274	0.774	1.0	0.991 to 1.01	0.938	0.971	0.998	0.987 to 1.01	0.697	0.494	0.996	0.986 to 1.005	0.356

Hop for distance	cm	274	0.194	1.001	0.99 to 1.013	0.863	0.979	1.004	0.99 to 1.018	0.589	0.238	1.009	0.997 to 1.021	0.131
			Multivariable negative binomial regression[c]				Multivariable negative binomial regression[c]							
		Hips[a]	*sex[b]	aIRR	95%CI	p	*sex[b]	aIRR	95%CI	p				
Variables	Unit	n	p	Superolateral defect severity (1-4)			p	Superomedial defect severity (1-4)						
Abduction	**10 N**	274	0.557	1.014	0.963 to 1.067	0.603	0.084	1.001	0.92 to 1.326	0.978				
Adduction	**10 N**	274	0.795	1.034	0.985 to 1.085	0.177	0.493	1.042	0.978 to 1.111	0.202				
Extension	**10 N**	274	0.072	**1.032**	**1.01 to 1.055**	**0.005**	0.461	1.01	0.992 to 1.029	0.289				
External rotation	**10 N**	274	0.251	1.012	0.947 to 1.084	0.711	0.409	1.014	0.916 to 1.121	0.793				
Internal rotation	**10 N**	274	0.202	1.019	0.947 to 1.096	0.621	0.707	0.947	0.842 to 1.065	0.361				
Flexion	**10 N**	274	0.953	1.02	0.994 to 1.046	0.135	0.305	1.018	0.979 to 1.059	0.373				
One-leg rise	reps	274	0.913	1.0	0.987 to 1.014	0.964	0.589	0.978	0.956 to 1.0	0.054				
Side-bridge	sec	274	0.495	1.028	0.997 to 1.009	0.352	0.322	0.999	0.99 to 1.008	0.766				
Hop for distance	cm	274	0.127	0.999	0.993 to 1.006	0.788	0.557	1.007	0.996 to 1.018	0.207				

Bold values indicate a statistically significant interaction or association (p<0.05)

For exposure variables (raw score in 10 N, reps, sec, cm), an OR/IRR >1 represents increased odds of cartilage defect presence or severity, respectively. For example, a 10 N (approximately 1 kg) increase in extension strength could be interpreted as having a 3% increased odds of worse superolateral hip cartilage defect severity.

[a]Number of hips with radiological alpha angle data.

[b]Strength/function variable*sex interaction.

[c]Adjusted for sex, test site, age, body mass index, alpha angle >60°.

cm=Centimetres, CI=Confidence interval, aIRR=Adjusted incidence rate ration, N-Newton, aOR=Adjusted odds ratio, reps=Repetitions, sec=Second.

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